Golla Sathish

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Profile Summary

Aspiring Data Scientist with hands-on experience in Python, SQL, machine learning, deep learning, and NLP through academic and self-driven projects. Skilled in building classification and regression models, data preprocessing, visualization, and model evaluation. Seeking an entry-level Data Science role to apply practical skills to data-driven problem solving.

Technical Skills

Programming: Python, SQL
Data Science: Pandas, NumPy, Data Cleaning, EDA, Data Visualization, Feature Engineering
Machine Learning: Linear/Ridge/Lasso/Logistic Regression, Decision Tree, Random Forest, Naive Bayes, XGBoost, LightGBM
Unsupervised ML: K-Means, Hierarchical Clustering, DBSCAN, PCA
Deep Learning: PyTorch, CNN, RNN, Transfer Learning, Computer Vision
NLP: Text Preprocessing, Tokenization, Stopword Removal, Stemming, Lemmatization, TF-IDF
Visualization: Matplotlib, Seaborn
Tools: Jupyter Notebook, VS Code, Streamlit, Git/GitHub

Projects

MNIST Digit Classifier *[Live Demo]* | *PyTorch, CNN, Streamlit* | Computer Vision / Deep Learning

- Built and trained a 2-layer CNN (206,922 parameters) in PyTorch on 60,000 MNIST images using Adam (lr 0.001, batch size 64); training loss fell from 0.23 to 0.02 in 5 epochs.
- Achieved 98.72% accuracy on the 10,000-image held-out test set.
- Deployed the saved model as a live Streamlit web app where users upload a handwritten digit image and get the predicted digit (0–9).

Student Performance Prediction App *[Live Demo]* | *Scikit-learn, Streamlit* | Machine Learning

- Built a Logistic Regression classifier on 395 student records to predict pass/fail from first- and second-period grades, using a stratified 80/20 split (316 train / 79 test).
- Achieved 87.3% test accuracy (vs. 67% majority-class baseline) and caught 24 of 26 failing students (92% recall).
- Excluded the final grade to prevent target leakage, and deployed the model as an interactive Streamlit app with sliders for real-time predictions.

RAG with Cross-Encoder Reranking *[Live Demo]* | *LangChain, FAISS, Streamlit* | Generative AI / NLP

- Built a two-stage RAG pipeline with LangChain: split a 720-word document into 18 chunks (500 characters, 100 overlap), embedded with MiniLM-L6-v2, and indexed in FAISS (top-20 retrieval).
- Added a cross-encoder reranker (ms-marco-MiniLM-L-6-v2) that rescores each question–chunk pair and keeps the top 5, cutting the context sent to the LLM by 72% (5 of 18 chunks).
- Generated answers with gpt-oss-120b (Groq API) from the retrieved context only; deployed as a live Streamlit app.

Customer Segmentation & Retention Prediction *[Live Demo]* | *Python, Pandas, Scikit-learn, K-Means, Logistic Regression, Streamlit* | Machine Learning

- Cleaned and analysed 805K+ retail transactions to understand customer purchasing behaviour and identify sales patterns.
- Performed RFM analysis on 5,878 customers and identified six customer segments, including Champions, Loyal Customers, and At Risk.
- Applied K-Means clustering on standardised RFM features to identify four groups based on customer behaviour.
- Built and compared Logistic Regression and Random Forest models to predict customer inactivity over a 90-day period.
- Developed and deployed a Streamlit dashboard to explore customer segments, inactivity risk, and generate predictions using the saved ML model.

Education

Sri Venkatesa Perumal College of Engineering and Technology — Tirupati, AP
B.Tech in Computer Science and Engineering | CGPA: 7.0/10.0 — *Sep 2022 – Sep 2026*

- Relevant Coursework: Python Programming, Data Structures, DBMS, Machine Learning

Sri Lakshmi Narasimha Senior College
Intermediate (Class XII) | CGPA: 6.1/10.0

Additional Information

Languages: English, Telugu (Native), Kannada
Interests: Data Science, Machine Learning, NLP, Problem Solving